Kenya Transport and Logistics Platform

Complete project architecture and operating blueprint

Passenger transport, goods delivery, freight, fleet operations and the systems that connect them.

Prepared 18 September 2026 | Version 1.0

The proposed platform supports passengers and goods through one shared business foundation, with separate workflows for seats, ride requests, shipments and vehicle operations. Customers use a common account; drivers, conductors, operators, branch staff and finance teams receive interfaces suited to their responsibilities.

This document explains the full target system: what each part does, how information and money move, how the apps connect to the network, and how a development team can implement and test it. A phased delivery plan covers build order while preserving the complete product scope.

Design decisions

• Support both company-owned vehicles and independent operators through an explicit vehicle ownership and operator model.

• Separate scheduled passenger booking, on-demand rides and goods transport while sharing identity, payments, fleet availability and tracking.

• Begin with a modular backend and managed infrastructure; extract independent services only when measured load or team ownership requires it.

• Treat booking inventory, money records, package custody and offline synchronization as core correctness requirements.

How to read this document

Business owners should begin with sections 1 to 5 and the implementation plan. Product designers should use the interface and workflow sections. Developers should use the module, network, data, integration and acceptance-test sections. Operational staff should review exceptions, settlements, hubs and support procedures.

The architecture is a proposed design. Vendor examples are based on published features, not access to their private architecture. Numeric performance targets and delivery estimates below are planning assumptions to validate during discovery and testing. Country-specific operating permissions and tax treatment must be confirmed for the selected business model.

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Figures are embedded beside their explanations. Reference numbers such as [4] connect factual claims to clickable sources in section 26.

1 Full product scope

The target platform is a transport marketplace and operations system. It can serve a single transport company first and support several operators later without changing the core meaning of a booking, trip, vehicle or payment.

Service	Customer purchase	Special capability
Scheduled buses and shuttles	Seat on a departure	Seat maps, segment inventory, boarding and manifests
On-demand rides	Pickup-to-destination journey	Driver matching, offer expiry, pickup PIN and trip sharing
Corporate and contract transport	Employee travel or recurring shuttle	Approvals, budgets, passenger rosters and monthly billing
Charter and vehicle hire with driver	Vehicle for a period or itinerary	Availability calendar, quotation, deposit and overtime
Local courier delivery	Pickup and delivery of packages	Package labels, proof of collection and proof of delivery
Intercity parcels	Branch or door-to-door shipment	Hub transfers, line-haul manifests and collection verification
Freight and trucking	Full or partial vehicle capacity	Weight, volume, equipment, carrier offers and rate agreements
Warehousing and fulfillment	Storage and order handling	Bins, stock movements, picking, packing and dispatch
Returns and claims	Return or resolution of failed service	Reverse shipment, inspection, claim evidence and refund
Cross-border and multimodal extensions	Journey with partner-controlled legs	Document references, border milestones and partner integrations

Shared scope

All services share customer identity, operator profiles, fleet records, communications, support, audit history and financial controls. Prices and operational states remain specific to the service. A passenger seat is never treated as interchangeable with cargo weight.

Specialized extensions

Air, sea and rail legs can be represented as partner-operated shipment legs with external booking references and milestones. Direct carrier booking, customs filing and specialized dangerous-goods or temperature-controlled operations require dedicated integrations and operating procedures. The data model accommodates them; access to those services is not assumed.

2 Researched examples and lessons

The following products illustrate complementary capabilities. Published feature descriptions show useful patterns; they do not establish local commercial availability, integration access, pricing or service quality.

Example	Published capability	Design lesson
Little [1]	Mobility, delivery and corporate transport	One account can span transport services; corporate controls deserve their own workflow.
BuuPass [2]	Bus search and booking across operators	Model routes, departures, boarding locations and operator-specific terms.
Onfleet [3]	Dispatch, route planning and delivery evidence	Give drivers explicit tasks and require evidence before closing deliveries.
Samsara [4]	GPS, diagnostics, routing and fleet operations	Separate vehicle telemetry from passenger and shipment transactions.
Uber Freight [5]	Freight procurement, execution, visibility and payments	Carrier sourcing and settlement must be first-class workflows.
Oracle Transportation Management [6]	Shipment planning, equipment, costs and stop sequencing	Orders, shipment legs and vehicle trips need distinct records.
Fleetio [7]	Inspections and fleet maintenance workflows	Connect vehicle defects to work orders and dispatch availability.
Odoo Barcode [8]	Barcode-based inventory receipts and deliveries	Scan both items and storage locations to preserve movement history.

Build versus integrate

Build the customer experience, booking rules, dispatch policy, operator permissions, operational records and financial reconciliation that define your business. Integrate specialist capabilities such as payment processing, map data, SMS delivery and optional telematics. Consider integrating an established warehouse or maintenance system if its workflows already fit the operation.

Each integration should sit behind an adapter: a small module that translates your internal request into the provider format and translates its response back. This reduces the effect of provider changes and makes testing possible without making live payments or sending real customer messages.

3 Users and permissions

A person can hold more than one role, but every action must be authorized for the relevant organization, branch and assigned job. Signing in establishes identity; authorization determines what that person may view or change.

Role	Normal access	Restricted actions
Customer or sender	Own bookings, receipts, shipments and claims	Cannot change confirmed fares or delivery evidence
Recipient	Scoped shipment tracking and collection verification	Cannot browse sender history or unrelated shipments
Driver	Assigned trips, necessary contacts and own earnings	Cannot confirm provider payments or approve own refunds
Conductor	Departure manifest, ticket validation and boarding	Cannot alter another departure or bypass seat constraints
Branch or warehouse staff	Assigned branch scans, handovers and inventory	Cannot change recorded custody history
Dispatcher	Assignment, reassignment and exception handling	Cannot ignore vehicle blocks without authorized resolution
Operator administrator	Own vehicles, drivers, departures and statements	Cannot access another operator or platform-wide finances
Corporate travel manager	Own employees, budgets, approvals and invoices	Cannot access other organizations
Support agent	Relevant customer case and limited operational context	Refund limits and sensitive-document access are controlled
Finance officer	Collections, reconciliation, invoices and settlement	High-risk payouts need a separate approver
Platform administrator	Configuration and approved cross-operator operations	Sensitive access is logged; use MFA and limited privileges

Operator isolation

Include an operator or organization identifier in owned records and enforce it on every query, export, file request and live-tracking subscription. Do not rely on hiding buttons. A platform administrator crossing organization boundaries should provide a reason for sensitive actions.

Approval and audit

Use separate initiation and approval for material payouts, bank or payout-number changes, and large refunds. Record actor, time, reason, previous value, new value and correlation ID. Correct mistakes through new events or adjustments rather than deleting financial or custody history.

4 App screens and customer experience

Interface	Screens	Key behavior
Customer app and website	Home, Travel, Send goods, quotes, checkout, active journey, history, receipts, support	One account; separate forms for each service; clear pending and failed states.
Driver app	Availability, offers, trip list, stop details, navigation, verification, evidence, earnings, incidents	Large practical controls; offline task list; clear upload and synchronization status.
Conductor mode	Departure, manifest, scan ticket, board passenger, parcel manifest	Validate the correct departure and prevent repeated boarding.
Operator portal	Onboarding, documents, vehicles, drivers, departures, jobs, statements	Show approval status and blockers before work is published.
Staff dashboard	Live map, dispatch board, bookings, shipments, branches, exceptions, finance	Combine map and list views; prioritize unresolved operational issues.
Branch and warehouse	Receive, print label, scan, sort, store, pick, load, handover	Every movement identifies the item, location and responsible person.
Corporate portal	Employees, travel policy, approvals, rosters, cost centers, invoices	Separate traveler experience from company approval and billing.

Kenyan customer details

Accept familiar Kenyan phone input and normalize it to international format. Display amounts in KES, use East Africa Time, and support English and Kiswahili. Capture a map pin, named stage or branch, landmark and written directions; a street address alone may be insufficient.

Accessible and low-bandwidth use

Keep booking forms short and preserve entered details after a connection failure. Use readable text, labeled controls, high contrast and touch targets suitable for use outdoors. Compress photos, avoid unnecessary map downloads and provide essential status by SMS where configured. A secure recipient tracking page should work without app installation.

What the interface must explain

Before purchase show the operator, service, included baggage or cargo allowance, pickup or boarding location, price, cancellation terms and expected service window. During payment distinguish authorization requested, processing, confirmed and unresolved. During a journey display the age of the last tracking update.

5 System context and service boundaries

All interfaces use the same authenticated platform. External providers perform specific services; the platform remains responsible for its own booking and accounting records.

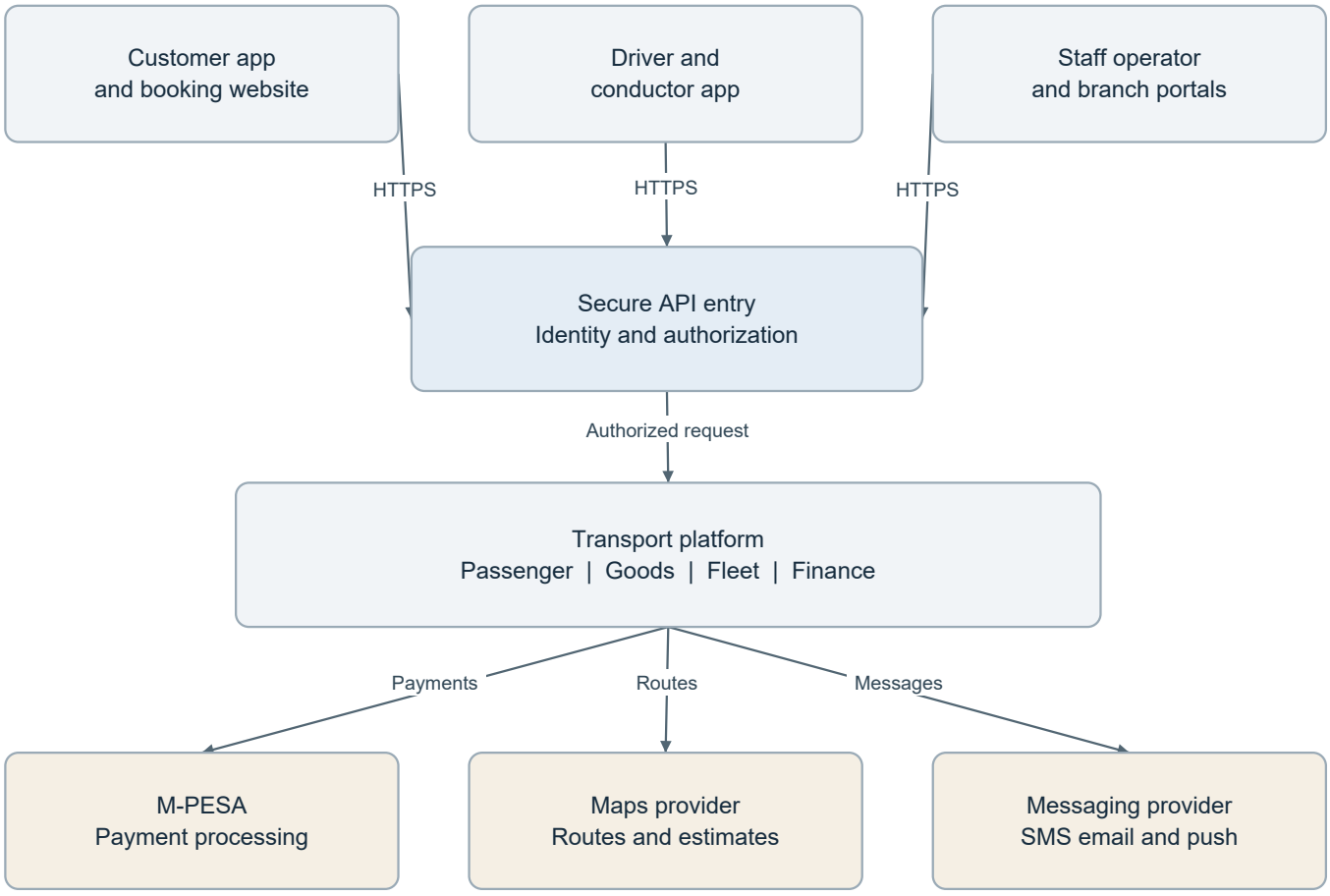


Figure 1 System context. Arrows show request direction; responses return over the same secured connection.

Important boundaries

A payment provider can report a collection result, but it cannot decide which seat should be assigned. A maps provider estimates travel, but it does not own the accepted fare. A notification announces a booking; it does not create the booking. These boundaries prevent provider failures from corrupting business records.

Customer tracking access is limited to the relevant journey and expires. Drivers see contacts only for assigned work. Staff actions pass through the same business checks as app actions, even where staff have broader permissions.

6 Backend modules and responsibilities

Use a modular monolith initially: one backend codebase with explicit module interfaces. Deploy the API and background workers separately. This preserves simple transactions while allowing slow work to run without blocking customer requests.

Module	Owns	Works with
Identity and organizations	Users, memberships, roles, operator approvals	All modules
Catalog and pricing	Services, routes, zones, rate cards, quote versions	Passenger, goods, corporate contracts
Passenger	Departures, seat holds, tickets, boarding, ride requests	Fleet, dispatch, payments
Goods and warehouse	Shipments, packages, custody, bins, stock movements	Trips, hubs, claims
Fleet and workforce	Vehicles, drivers, shifts, capacities, documents, maintenance	Dispatch and operator portal
Dispatch and execution	Assignments, trips, stops, offers, route versions	Passenger, goods, tracking
Payments and ledger	Payment attempts, refunds, balanced entries, settlement	Providers, orders, invoicing
Tracking and communication	Location history, latest position, notifications	Trips and authorized viewers
Support and reporting	Cases, incident workflows, derived business measures	Operational records and audit trail

Reliable background work

When a business change commits, save its event in an outbox table in the same database transaction. A relay publishes that event to a durable queue. Workers process it, record the event ID, and acknowledge it only after successful work. Retries must be safe because messages may be delivered more than once. [13]

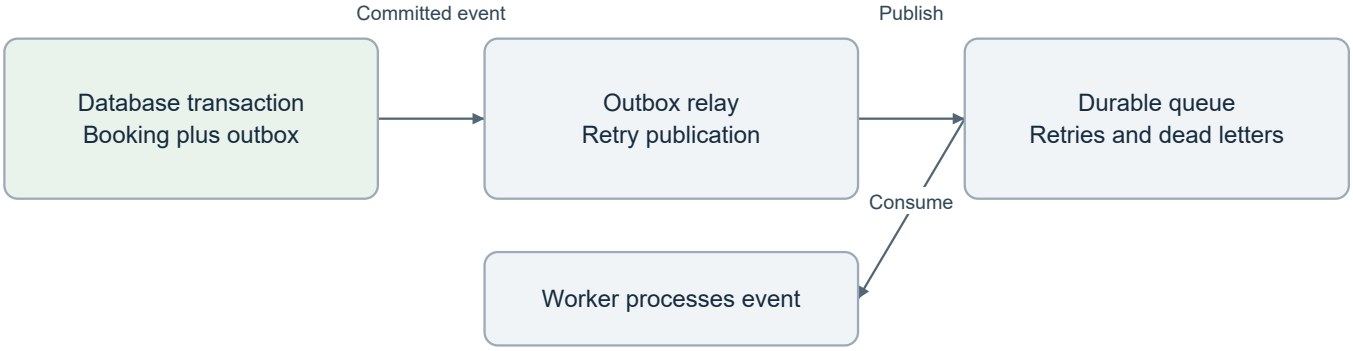


Figure 2 A committed booking survives even if notification delivery is temporarily unavailable.

7 Scheduled passenger booking

A route defines an ordered set of stops. A departure gives that route a date, time, operator, vehicle class and fare schedule. Seat inventory is held against the departure and occupied route segments.

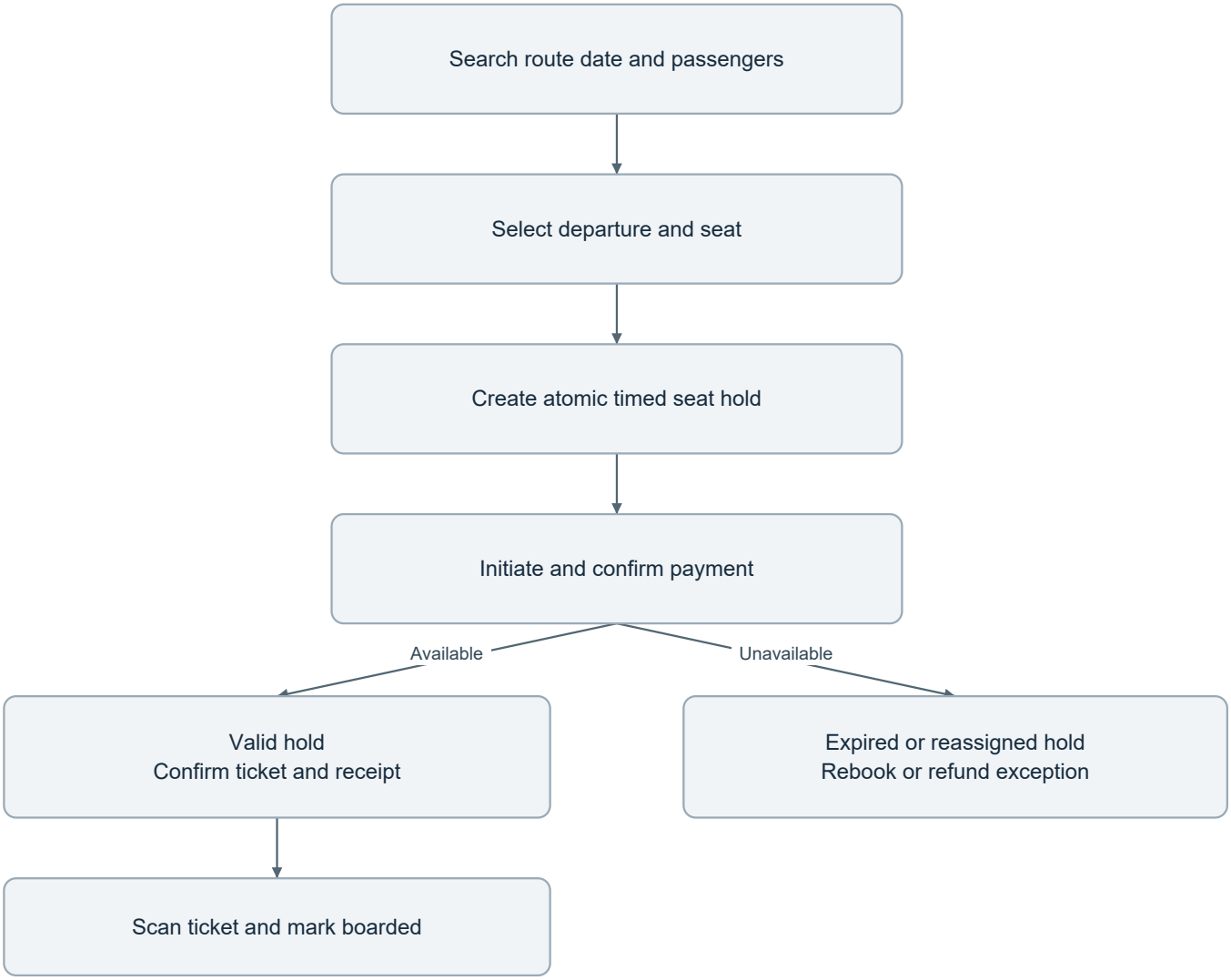


Figure 3 Scheduled booking. Payment success and seat availability are evaluated together before issuing a confirmed ticket.

The server enforces hold expiry and seat uniqueness within a transaction. It must reject overlapping segment reservations even if two customers click at the same instant. PostgreSQL supports row locking and transactional control for this purpose. [14]

For Nairobi to Nakuru to Eldoret, a seat may be sold to different passengers on non-overlapping segments. For group bookings, acquire the requested seats together or return a clear partial-availability response before payment.

Tickets contain a random identifier or signed token, departure reference, passenger details required for operation and boarding information. The scanner checks departure, ticket state and previous boarding. Offline manifest checks are possible, but disconnected scanners cannot guarantee immediate duplicate detection across devices.

Handle cancellations, rescheduling, vehicle substitution, seat-map changes, no-shows and delayed departures explicitly. A replacement vehicle with fewer seats creates an exception requiring reassignment and customer communication; it must never silently remove passengers.

8 On demand rides and corporate transport

On-demand ride workflow

The customer sets pickup and destination, receives a versioned quote and submits a ride request. Dispatch filters approved drivers by availability, vehicle class, service zone and document status. It sends time-limited offers. The first valid acceptance wins through an atomic assignment operation; later or expired acceptances are rejected.

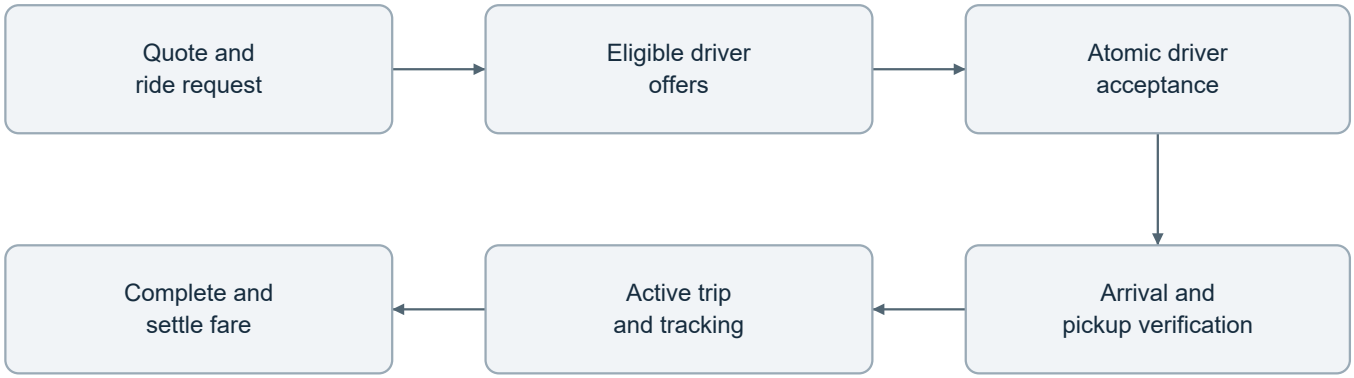


Figure 4 On-demand ride lifecycle. Offer acceptance and trip start are separate events.

Confirm pickup through an agreed method such as a trip PIN. Start time, route events and approved waiting rules support fare calculation. Destination changes require a revised estimate and customer acceptance where appropriate. Record cancellations and no-shows with reasons, evidence and the applicable price policy.

Corporate travel and recurring shuttles

A company defines eligible employees, cost centers, budgets, approval thresholds and permitted services. A travel request may require approval before dispatch. Recurring shuttles create dated departures from a schedule template and maintain a passenger roster. An employee leaving the company must lose future access without deleting historical invoices.

Charter and vehicle hire

A hire request specifies dates, itinerary, vehicle class, passenger count and included hours or distance. Staff issue a quote, reserve the vehicle and driver for the entire interval, record deposits, and finalize permitted extras at completion. Availability checks include travel to the pickup, turnaround time and existing reservations.

Safety and failed matching

Provide trip sharing, incident reporting and a documented escalation process. An emergency button must state its actual response path. If no driver accepts, show the customer a clear retry, scheduled alternative or cancellation outcome and release any reservation or authorization according to the payment policy.

9 Goods delivery and freight

A goods booking records sender, recipient, collection and destination, package count, dimensions, weight, declared value, handling needs and service window. Quotes use the applicable tariff; a business may charge by actual weight, volumetric weight, distance, zone or vehicle capacity. Save the accepted quote and measurement basis.

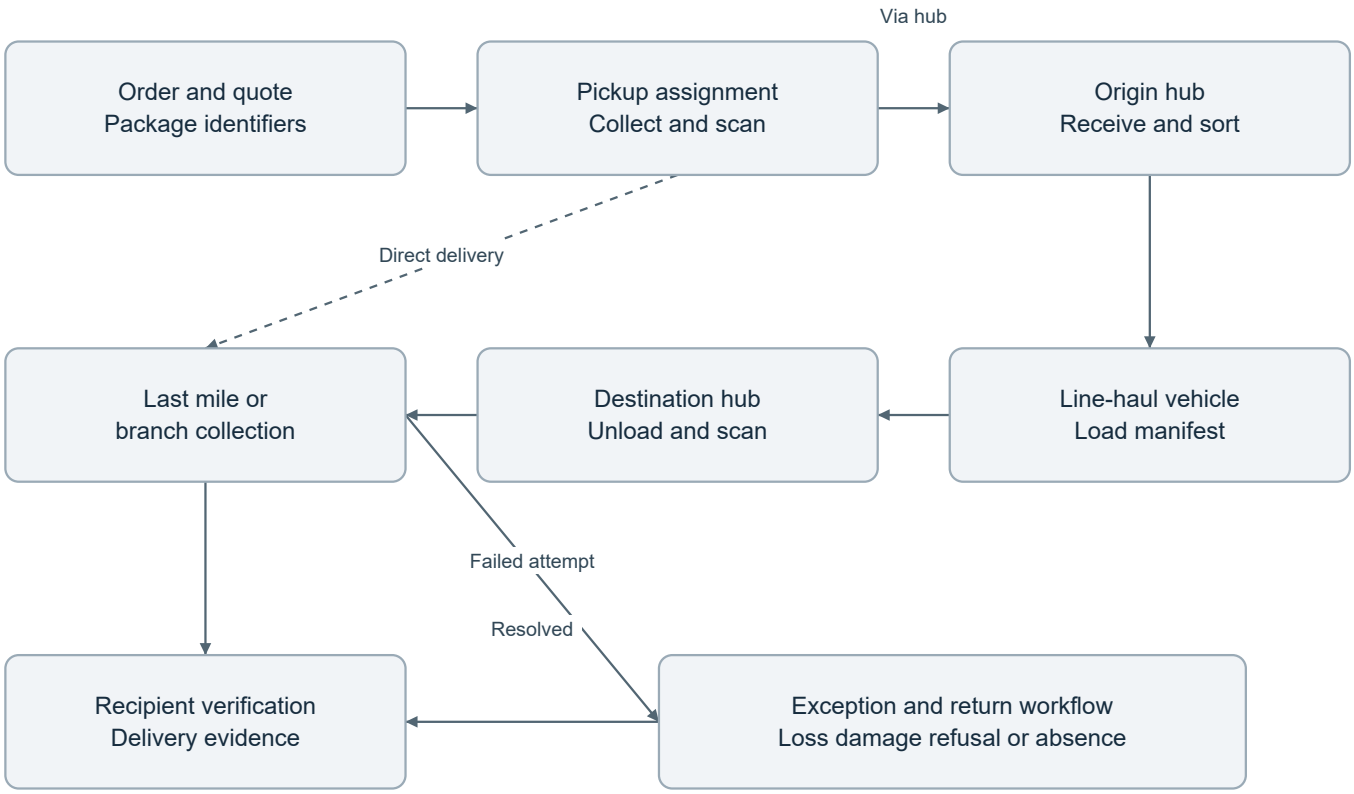


Figure 5 Goods movement. Direct deliveries skip hubs; intercity parcels retain a custody event at every transfer.

Freight procurement

For full-truckload or shared-load freight, include equipment type, loading access, weight and volume limits, loading dates, delivery windows and agreed charges. Staff may assign an owned truck, request carrier offers or select an approved contract rate. Preserve the accepted carrier offer and terms before dispatch.

Completion and exceptions

Capture collection evidence and delivery evidence separately. Evidence can include recipient OTP, signature, package scans, photos and delivery notes, depending on the service. Partial deliveries remain visible at package level. Failed attempts create an exception with the next action: retry, hold at branch, return or claim.

Returns create a linked reverse shipment with its own legs and custody history. Claims reference the original packages, declared value, condition evidence and responsible reviewers. Do not let a photograph alone prove that the intended recipient received the correct package.

10 Hubs warehouses and custody

Parcel hub operations

A hub is a physical handover point. On arrival, scan package and manifest, compare expected and received items, record condition, and place the item in a named zone or bin. A departure manifest lists the packages assigned to a trip. Loading scans must reconcile against the manifest before release.

A custody record identifies the package, giving party, receiving party, location, event type, device event time, server receipt time and evidence. Corrections add a new event with a reason. Missing, extra or damaged packages create exceptions rather than being silently accepted.

Warehouse inventory and fulfillment

Stored merchant stock requires a warehouse model beyond parcel tracking: owner, SKU, lot or serial where relevant, quantity, unit, bin and stock status. Goods receipt increases available stock after checks. A customer fulfillment order reserves stock, generates picking tasks, confirms packed quantities and creates a shipment at dispatch. Barcode-based receipts and deliveries are an established pattern. [8]

Movement	Record created	Control
Receive at branch	Package receipt and custody event	Compare against expected manifest
Store inventory	Stock movement into bin	Identify stock owner and quantity
Pick and pack	Reservation consumption and packing record	Prevent picking more than available
Load vehicle	Manifest scan and custody handover	Validate vehicle cargo limits
Transfer hub	Unload and discrepancy record	Reconcile expected versus received
Collect at branch	Recipient check and release event	Verify recipient or authorized delegate
Return or quarantine	Linked movement and exception	Exclude blocked stock from allocation

Mixed passenger and parcel operations

Passenger capacity and cargo capacity remain independent. Model seats, baggage allowance, cargo weight, volume and compartment restrictions. Only vehicles approved for the relevant services can receive mixed work. Parcel pickup windows must not override the published passenger timetable.

Special handling

Temperature-controlled items require sensor readings, agreed thresholds and an excursion response. Dangerous or restricted goods require explicit acceptance rules, handling procedures and the applicable permissions. These categories should be disabled until the business is prepared to operate them.

11 Dispatch routing and fleet availability

Dispatch connects demand to real resources. It decides who performs the work, in which vehicle, in what order and at what time. A map recommendation is advisory until the platform validates and commits the assignment.

Input	Why it matters
Job type and service promise	Passenger departure, pickup window and delivery deadline impose different constraints.
Vehicle capability	Seat count, cargo capacity, equipment and permitted use determine eligibility.
Driver approval and shift	Documents, skills, working availability and breaks affect assignment.
Existing work	Travel, service time and turnaround buffers prevent overlapping commitments.
Location and route estimate	Distance and estimated arrival affect allocation and customer promises.
Maintenance and incidents	Blocked or unsafe vehicles must be excluded.
Operator and contract	Approved carrier list, tariff and settlement terms control outsourced work.

Planning method

Start with dispatcher-assisted assignment and transparent rules. For delivery batches, a route optimizer may propose vehicle assignments and stop order subject to time windows, capacity and working hours. Google documents these constraints in its route optimization API. [11] Validate its proposed result against your own business rules before publication.

Save each route plan version. Once a driver begins a route, preserve completed stops and communicate any remaining-stop changes clearly. Do not automatically reorder scheduled passenger stops for parcel convenience. If no feasible plan exists, return an explanation to dispatch rather than hiding unassigned work.

Reassignment and breakdown

When a vehicle breaks down, block future allocation, open an incident, select a replacement, and create explicit transfers for passengers or packages. Record when the old assignment ends and the replacement starts. Notify affected customers with a revised estimate and preserve the original service promise for reporting.

Allocation correctness

Use a transaction and version check when accepting an offer or assigning a trip. Lock or constrain the relevant resource interval. A previously valid availability screen is not proof that the vehicle is still free when the assignment reaches the server.

12 Fleet maintenance and tracking hardware

Fleet management controls the readiness and cost of vehicles. It includes registration, ownership, operator, vehicle class, seating layout, cargo limits, mileage, inspections, service history, fuel and incidents.

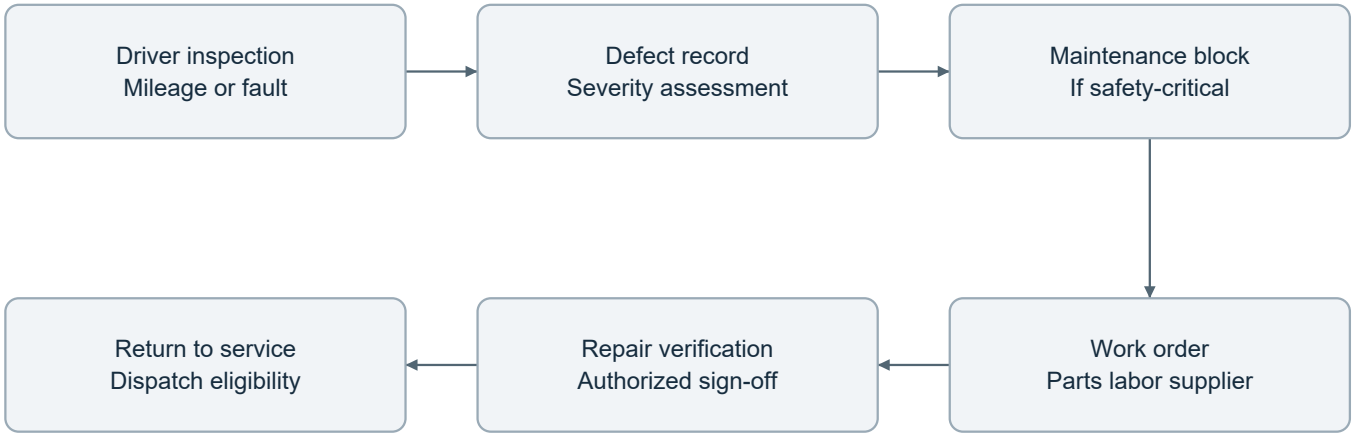


Figure 6 A reported defect can directly affect dispatch availability. Maintenance release requires verification.

Maintenance and expenses

Create service reminders by date, distance or operating hours as appropriate. Record inspections, defects, work orders, parts usage, suppliers, labor and downtime. Fuel entries include quantity, cost, odometer, vehicle, driver and receipt. Flag unusual readings for review rather than treating them as automatic fraud. Fleetio and Samsara illustrate these maintenance and telemetry patterns. [4, 7]

Phone tracking and installed trackers

Phone GPS is useful for driver workflow and active-trip tracking. An installed tracker can report vehicle movement independently of the driver phone and may provide ignition, mileage or diagnostic data. Capabilities depend on the actual device and provider; do not assume that every tracker supplies fuel or engine information.

Prefer a provider API adapter for the first hardware integration. If ingesting devices directly, use a separate device gateway, per-device credentials, protocol validation, bounded message sizes and rate limits. MQTT over TLS is one possible protocol where supported; device-specific TCP protocols require an isolated ingestion service.

Store source identity and quality flags with telemetry. Define which source is preferred when phone and tracker disagree, and show uncertainty instead of mixing both into a misleading continuous path.

13 Payments refunds and settlement

Collections, accounting and payouts are separate concerns. Safaricom publishes APIs for customer payments, transaction queries and business payments; production capabilities depend on approved onboarding and enabled products. [9]

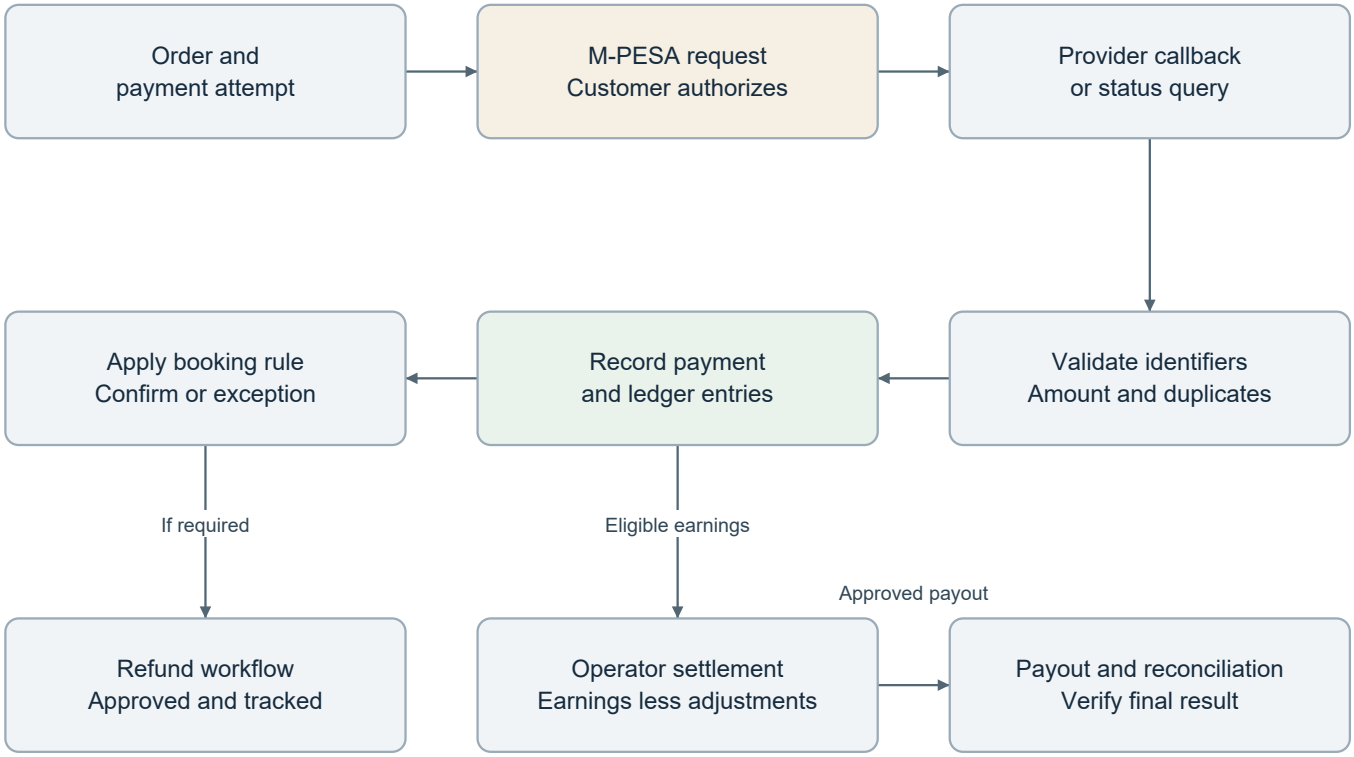


Figure 7 Money flow. Provider acknowledgements, customer payments and operator payouts are distinct records.

Payment attempts use pending, successful, failed or unknown states. A timeout does not prove failure. Correlate provider references, check expected amount and currency, apply available authenticity controls and deduplicate callbacks. Reconcile unknown results before retrying money movement.

Use an append-only, balanced ledger with exact monetary amounts. Record collections, operator liabilities, platform fees, refunds, cash handling and adjustments separately. Account mapping depends on whether the business acts as principal or agent and must be approved by its accountant.

Settle only eligible earnings after the agreed completion and dispute conditions. Use separate payout initiation and approval. Record each payout attempt and final provider result. Prevent duplicate payouts after timeouts. Cash collected by a driver remains a separate obligation until remitted and reconciled.

Keep tax invoicing separate from payment receipts. KRA offers eTIMS system-to-system integration; select the appropriate approved integration and tax treatment for the business. [10]

14 Tracking and offline operation

The driver app keeps a local copy of assigned work and a durable queue of pending actions. This follows the local-plus-network data pattern documented by Android. Background location collection is constrained by operating-system policy and battery behavior. [12]

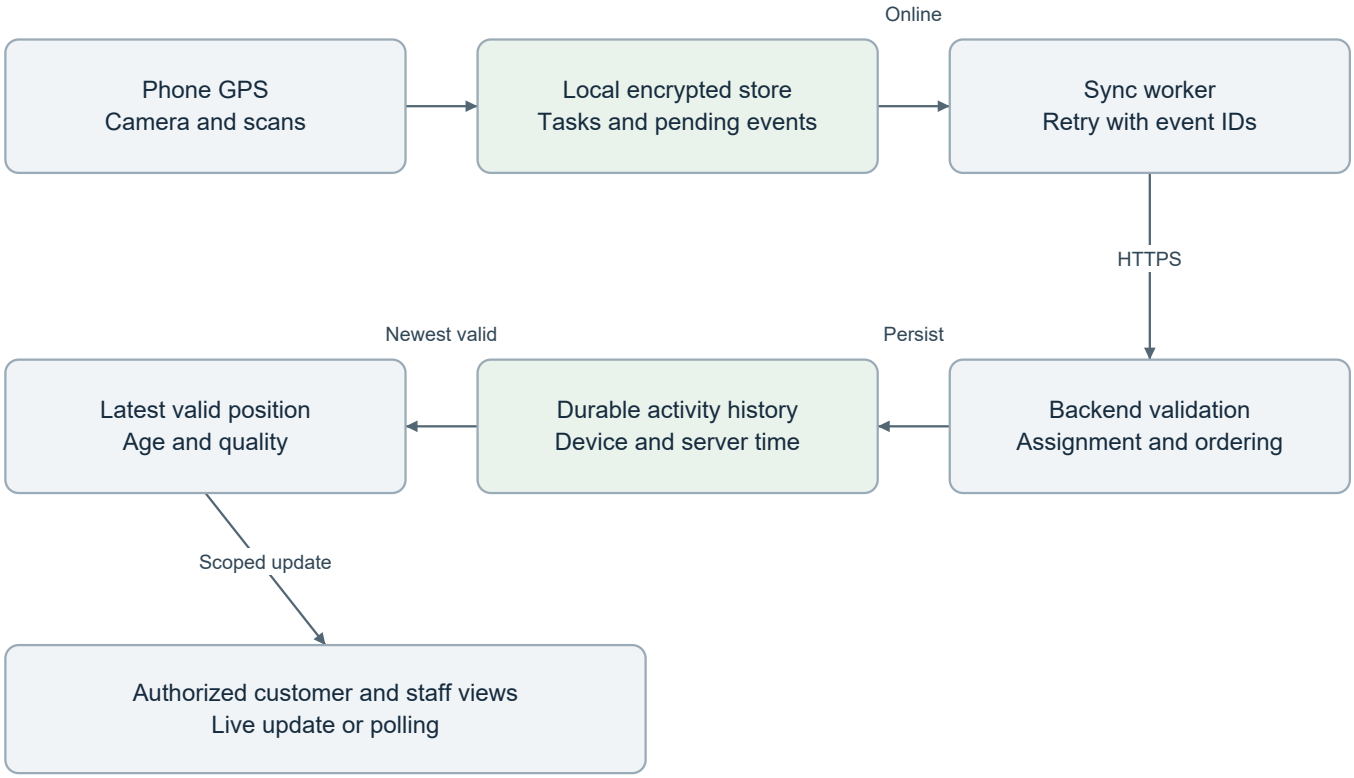


Figure 8 Offline synchronization and tracking. Old samples remain historical and cannot replace newer current positions.

Synchronization rules

Every event contains a unique ID, actor, device, assignment, entity version, capture time and payload. The server validates permissions and allowed transitions, then returns accepted, duplicate or rejected. The device removes a queued event only after acknowledgement. Rejected actions remain visible for resolution.

A driver can capture delivery evidence offline; the interface shows pending synchronization. New seat confirmation, provider payment confirmation, payout and resource assignment require server authority. For large photos, obtain a short-lived upload permission and retry the upload independently from the event metadata.

Tracking behavior

Configure sample intervals by motion and trip state. During testing, evaluate a proposed 10 to 30 second active-trip interval, but do not promise it on every device. Show the last update time, accuracy and stale status. Stop or reduce collection outside authorized work according to the tracking policy. Avoid exposing a driver's entire route history to a recipient.

15 Cloud network and office connections

Public endpoints accept customer traffic and provider callbacks. Business services and data stores sit in a private network. Network restrictions complement application authorization; both are required.

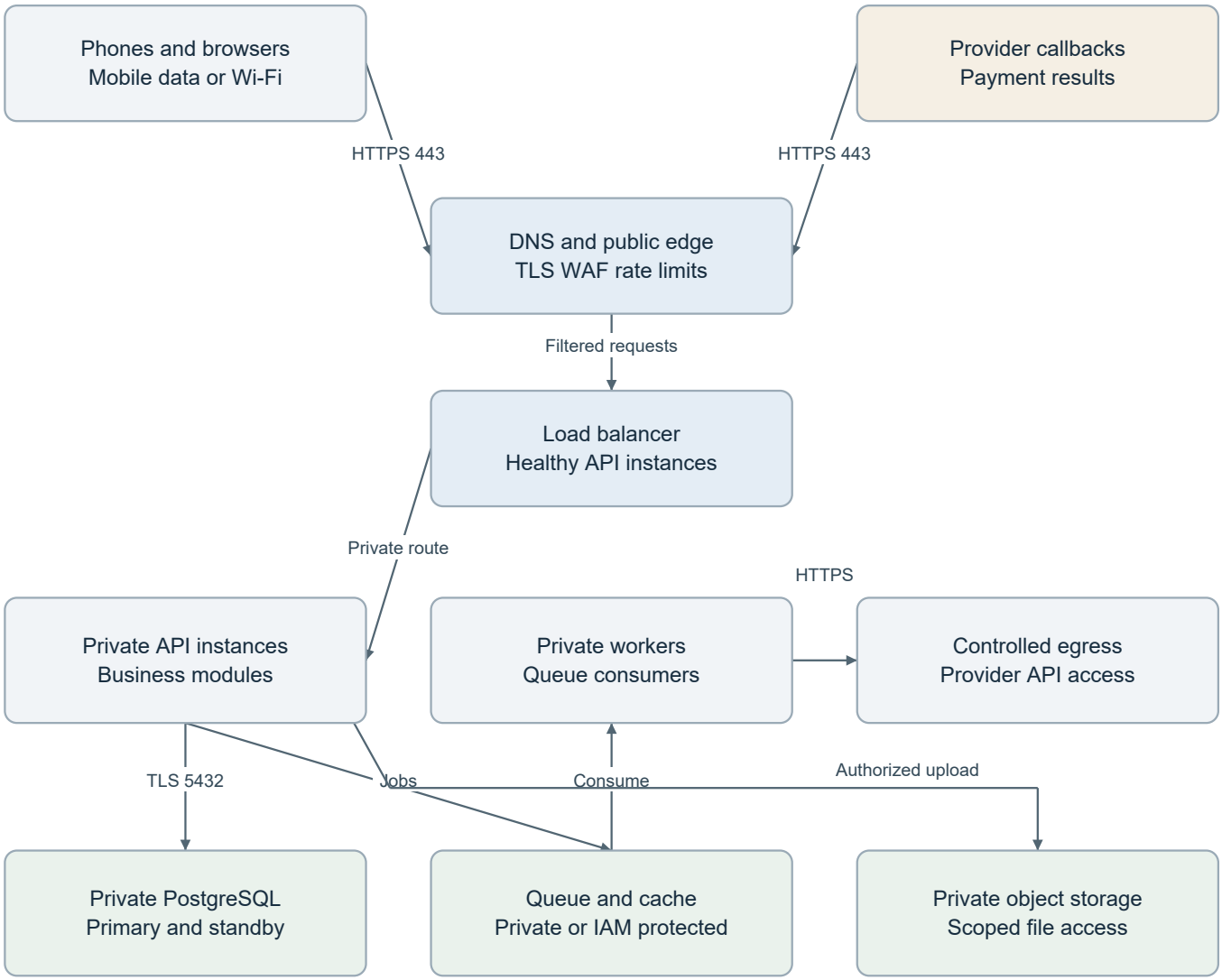


Figure 9 Deployment network. Return traffic is implicit. Workers also access the database through restricted credentials; the API uses controlled outbound access for synchronous provider calls.

HTTPS encrypts ordinary requests; WSS supports live updates. The load balancer routes to healthy instances. Private database access is restricted to required service identities. Storage rejects public reads; authorized apps use short-lived object-specific URLs when needed. Cache and queue ports depend on the chosen managed service and must not be exposed publicly.

Office and branch devices can use normal internet access. Separate staff and guest Wi-Fi, protect devices, and use backup connectivity for critical branches. Administrative maintenance uses MFA, least privilege and audited sessions. Separate development, staging and production accounts or projects and secrets.

16 Database relationships

This conceptual diagram shows ownership and reference relationships, not every table or foreign key. One customer order has a service type and links to its passenger or goods detail. A physical trip can carry several shipments where the vehicle and service permit it.

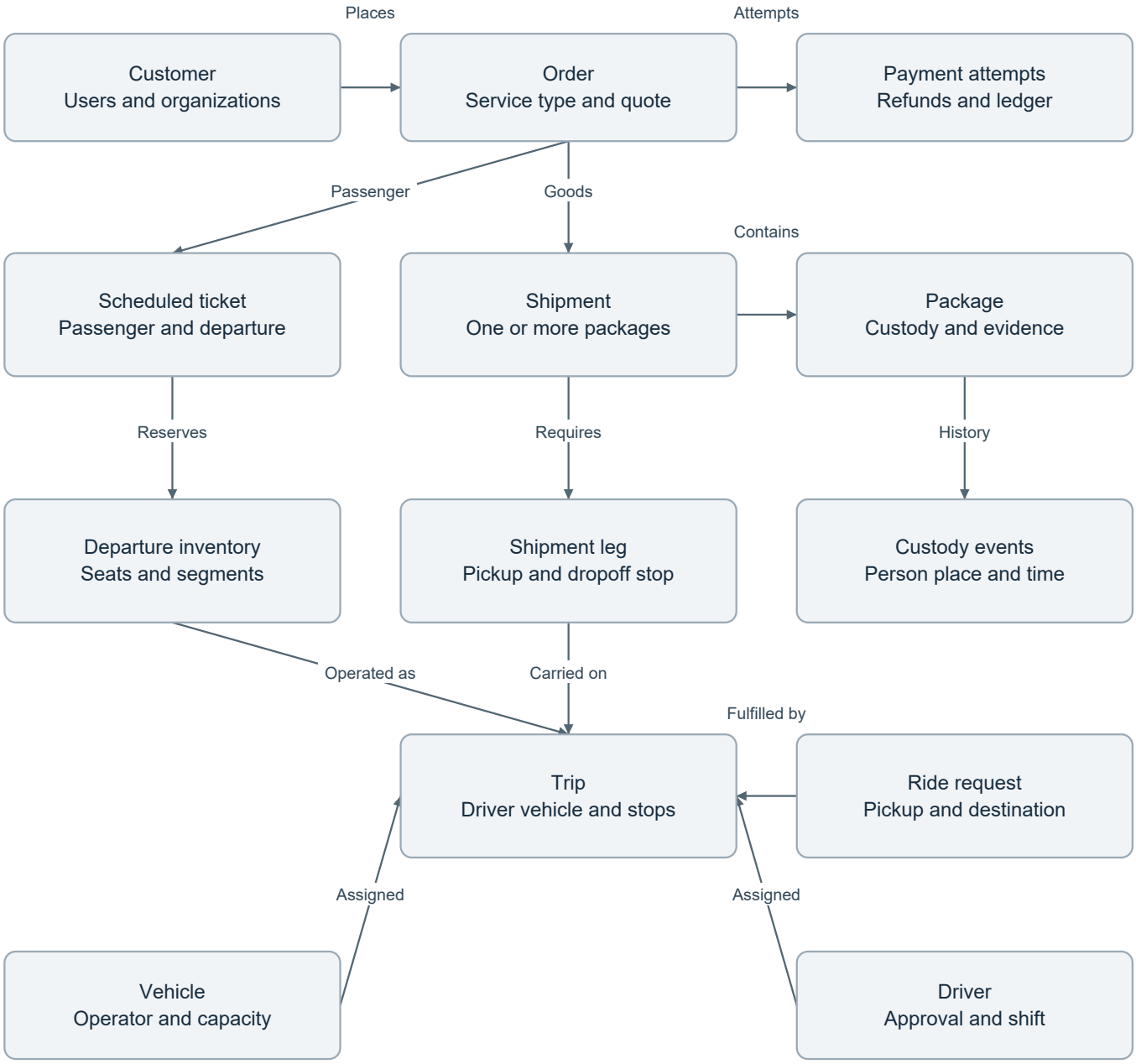


Figure 10 Core data relationships. Orders also link to their ride requests. Shipment legs connect freight to actual vehicle trips without losing package identity.

Additional tables cover role memberships, operator documents, addresses, rate cards, stock movements, inspections, work orders, support cases, notifications, audit events, API idempotency keys and the transactional outbox.

Use foreign keys and database constraints where possible. Keep accepted quotes and historical operator agreements immutable. High-volume telemetry can move to partitioned tables or a dedicated time-series store as actual volume grows.

17 Data dictionary and business constraints

Record	Minimum important fields	Invariant
Order	ID, customer, service type, currency, quote version, status	Exactly the permitted service-specific details
Departure	Operator, route, stop times, vehicle class, seat layout version	Published inventory remains consistent with sold tickets
Seat reservation	Departure, seat, segment, hold expiry, booking	No active overlapping occupancy
Ride request	Pickup, destination, quote, offer state, assignment	Only one winning active driver assignment
Package	Shipment, label ID, weight, size, handling flags	Unique identifier retained through all legs
Trip and stop	Vehicle, driver, planned times, actual times, sequence	No incompatible resource overlaps
Custody event	Package, actor, location, type, times, evidence	Append history; corrections are new records
Payment attempt	Order, expected amount, provider request/reference, state	Deduplicate provider transaction and callback
Ledger entry	Journal, account, amount, currency, order reference	Balanced journal with immutable posted entries
Settlement	Operator, period, eligible items, adjustments, payout state	No duplicate settlement of the same entitlement
Tracking sample	Source, trip, position, accuracy, event and receipt time	Historical samples cannot move current state backwards

State is more than a single status

Keep commercial, operational and payment state separate. A shipment can be delivered while payment is unresolved. A ticket can be paid but cancelled. An operator can have earned an amount that has not yet been settled. Collapsing these into one status produces ambiguous reports and unsafe actions.

Consistency and retention

Use exact decimal or minor-unit money storage, timezone-aware timestamps and explicit units for weight and distance. Define field validation and maximum sizes. Use optimistic versions for edits and transactions for scarce inventory. Retention is category-specific: operational GPS, identity documents, financial records and incident evidence should not share one arbitrary deletion period.

18 APIs events and external integrations

An API is the controlled interface used by apps and partners. Use versioned endpoints, authenticated calls, explicit permissions and consistent errors. The following names are proposed contracts, not existing provider APIs.

Operation	Proposed endpoint	Main checks
Price request	POST /v1/quotes	Service, locations, capacity, tariff version
Seat hold	POST /v1/seat-holds	Departure, segment availability, expiry
Order creation	POST /v1/orders	Quote validity, customer and service details
Request ride	POST /v1/rides	Service zone and quote
Accept offer	POST /v1/offers/{id}/accept	Driver, expiry and atomic winning assignment
Start payment	POST /v1/orders/{id}/payments	Idempotency key and expected amount
Sync driver events	POST /v1/sync/events	Unique IDs, assignment, version and state
Location batch	POST /v1/trips/{id}/locations	Driver/source, quality and bounded batch
Delivery evidence	POST /v1/packages/{id}/proof	Authorized actor and required evidence
Provider callback	POST /v1/webhooks/{provider}	Provider controls, correlation and deduplication

Event contract

Examples include BookingConfirmed, DriverAssigned, PackageCollected, PackageTransferred, DeliveryCompleted, PaymentSucceeded and SettlementApproved. Include event ID, schema version, organization, entity ID, occurred time and correlation ID. Minimize personal data. Consumers record processed IDs; retries should reproduce the same business result.

Integration inventory

Required launch adapters: M-PESA, maps, SMS or push, identity/session handling and file storage. Additional adapters: eTIMS, accounting, corporate ERP, e-commerce orders, telematics, warehouse management and partner carriers. An integration contract should define credential owner, callback handling, timeouts, retries, quotas, monitoring and reconciliation responsibility.

Keep secret provider credentials on the backend. Restrict client map keys using the provider-supported application restrictions. Provide partners with scoped keys or tokens, rate limits and signed outbound webhooks where supported. Retry partner delivery with a stable event identifier and publish a replay procedure.

19 Security privacy and Kenyan requirements

Identity and access

Use short-lived sessions or access tokens, secure refresh-token storage, logout/revocation and MFA for privileged users. Rate-limit OTP and login requests. Verify object ownership on every API request. Validate authorization again for files, exports and live-tracking subscriptions. Do not log passwords, access tokens, full identity documents or payment secrets.

Application and infrastructure controls

Encrypt network traffic and protected stored data. Keep secrets in a managed secret store. Scan uploaded files, limit file types and sizes, and quarantine documents until checks pass. Apply dependency updates, code review and environment separation. Audit administrator access and protect backups with separate credentials.

Location and identity data

Explain when tracking runs and who can view it. Limit recipient access to the relevant active shipment and avoid exposing unrelated stops. Collect only necessary identity and passenger details. Establish retention, access-request, correction and deletion processes, with approved exceptions for records that must be retained.

Kenyan operational readiness

ODPC identifies transport services, including online passenger-hailing applications, within registration categories. Determine the platform and operator controller/processor roles and applicable registration obligations. Review the need for a data protection impact assessment for the chosen processing, particularly systematic tracking. [17]

Kenya has specific transport-network-company regulations. Confirm the rules and permissions that apply separately to ride-hailing, scheduled passenger services, goods carriage and partner operators. Track document types, issuer, reference, approval, expiry and verification history in the platform; uploading a file is not verification. [18]

Evaluate cross-border data transfers and hosting arrangements before selecting production regions. Decide who issues tax invoices and how eTIMS applies to the business. Do not hard-code commission limits, tax rates or regulatory thresholds without checking the current rules and effective dates. [10]

Incident response

Maintain a named response owner, escalation contacts, containment procedure and evidence-preservation process. Separate operational incidents from cybersecurity incidents while linking related cases. An emergency feature requires trained people and tested procedures; a button alone is not an emergency service.

20 Reliability monitoring and recovery

These are proposed engineering targets for budgeting and testing, not commitments already achieved. Final service levels depend on operating hours, provider dependencies and support staffing.

Area	Initial planning target	Validation
Core API availability	99.9% monthly target	Synthetic booking checks and incident records
Ordinary API latency	95% under 1 second excluding external provider delay	Load tests with representative database volume
Recovery point	Up to 15 minutes of data exposure as a planning ceiling	Verify actual backup and transaction-log configuration
Recovery time	Restore essential service within 4 hours	Timed recovery exercise
Tracking freshness	Show stale warning after a configured interval	Real device and poor-network field tests
Financial correctness	No duplicate charge recording or payout	Retry, timeout and reconciliation scenarios

Failure handling

Database outage: stop new authoritative confirmations and show a clear pending/unavailable state. Payment delay: preserve the attempt as unresolved and reconcile. Maps outage: retain valid existing plans and use a controlled manual dispatch path. Messaging outage: keep the booking valid and retry notifications. Worker failure: retry with backoff and move persistent failures to a dead-letter queue with an alert.

Operational observability

Collect structured logs, metrics and traces with a shared correlation ID. Monitor API errors and latency, queue age, failed jobs, payment mismatches, stale GPS, database capacity, failed sign-ins and notification delivery. Create alerts with an owner and a practical response procedure; avoid alarms with no action.

Recovery and deployments

Use encrypted backups, point-in-time recovery where available, and restoration drills. Keep a second protected backup location according to approved residency rules. A database standby improves availability but does not replace backups. Use backward-compatible schema changes, health checks, staged releases and a tested rollback strategy.

Capacity example

If 1,000 active vehicles each send a location every 15 seconds, ingestion is about 67 samples per second and 1.92 million samples during an eight-hour day. This is illustrative, not a demand forecast. Measure payload size, index cost, retention and map usage before selecting infrastructure.

21 Reporting and business measures

Reports should trace back to operational and financial events. Define each measure before building a dashboard so different teams calculate it consistently.

Measure	Definition	Primary user
Booking conversion	Confirmed bookings divided by eligible started booking sessions	Product and marketing
Seat utilization	Occupied seat-segments divided by offered seat-segments	Passenger operations
Pickup punctuality	Pickups within the agreed window divided by completed pickups	Dispatch
Delivery success	Delivered packages divided by packages due in the reporting cohort	Goods operations
First-attempt delivery	Packages delivered on first attempt divided by attempted packages	Last-mile manager
Vehicle utilization	Productive assigned hours divided by available operating hours	Fleet manager
Empty running	Distance without the defined revenue load divided by measured distance	Freight and fleet
Payment exceptions	Unresolved collections by age and amount	Finance
Operator payable	Earned eligible amounts less settled amounts and valid adjustments	Finance and operators
Contribution per service	Recognized service income less agreed direct service costs	Management
Custody discrepancies	Missing, extra or damaged items per handover cohort	Hub manager
Support resolution	Time from case creation to agreed resolution state	Support lead

Dashboard views

The dispatcher sees unassigned work, active trips, delays and incidents. Finance sees pending payments, mismatches, refunds and payout approvals. Operators see their own utilization and earnings. Management sees trends by route, service, branch and operator. Customers see only their journeys and receipts.

Analytics architecture

Begin with indexed reporting queries and scheduled summaries that do not burden booking transactions. As reporting grows, feed a separate analytics store from committed events or controlled extracts. Mark data freshness and protect personal information in exports. Reconcile finance reports to ledger totals rather than inferring revenue from completed-trip counts.

22 Technology and code organization

Layer	Proposed starting choice	Purpose
Customer and driver apps	Flutter	Separate apps sharing selected models and UI components [15]
Staff and operator web	React with TypeScript	Forms, tables, dispatch and administration
Backend	NestJS with TypeScript	Explicit business modules and API contracts [16]
Database	Managed PostgreSQL with optional PostGIS	Transactional records and geographic queries
Local mobile storage	Encrypted database and secure credential store	Assigned work and pending offline events
Cache	Redis when needed	Latest positions and disposable acceleration
Jobs	Managed durable queue and workers	Retries, notifications and provider reconciliation
Files	Private object storage	Documents, photos and evidence
Maps and routing	Provider adapter; evaluate Google Maps	Map display, estimates and optional optimization [11]
Operations	Managed containers or equivalent app hosting	Repeatable deployment without premature platform complexity

Suggested repository structure

Keep apps/customer, apps/driver, apps/operations-web and apps/operator-web for interfaces. Keep services/api and services/worker for runtime processes. Within the API, separate identity, pricing, passenger, goods, warehouse, fleet, dispatch, payments, tracking and support modules. Shared packages hold API schemas, common types and design components; infrastructure definitions and operational runbooks live in dedicated directories.

Development rules

Each module owns its records and exposes explicit operations. Avoid a shared folder of business functions that any module can modify. Document API schemas and validate incoming payloads. Use database migrations, seeded non-production data and provider test environments. Never copy production personal data into development by default.

When to split services

Extract tracking ingestion when telemetry load or storage differs materially from bookings. Extract route optimization if it needs independent compute. Separate finance deployment only with a clear transaction and reconciliation design. A microservice split must solve an observed scaling or ownership problem; every additional service introduces network failure and operational work.

23 Testing and acceptance criteria

Scenario	Required result
Two buyers select the last seat	Only one overlapping reservation succeeds; the other sees accurate availability.
Two drivers accept one offer	Only one assignment is committed and both apps receive a clear result.
Payment succeeds after hold expiry	Payment is recorded; ticket issues only if inventory can still be secured, otherwise exception handling starts.
Callback repeats or arrives out of order	One financial effect; stale events cannot reverse a newer authoritative state.
Payout request times out	No blind duplicate payout; provider status is resolved or escalated.
Driver syncs delivery twice	One delivery event and one notification intent, with duplicate acknowledged safely.
One of three packages is missing	Partial completion is visible and the missing package remains open.
Driver is offline during boarding	Cached policy applies; pending checks and duplicate risks are explicit.
Vehicle breaks down	Future allocations block; replacement and custody transfers are recorded.
Operator changes a URL or record ID	Cross-operator data remains inaccessible, including files and live updates.
Database restored from backup	Restore meets tested objectives and financial reconciliation identifies any gaps.
App upgrade meets old pending events	Supported event versions synchronize or return actionable errors without data loss.

Test layers

Use unit tests for pricing and state rules, database integration tests for inventory and ledger invariants, provider-contract tests for callbacks, and end-to-end tests for complete journeys. Test Android devices with background restrictions, low battery, airplane mode and intermittent mobile data. Use security testing for authorization, uploads and privileged actions.

Pilot acceptance

Run real operational rehearsals with controlled payments, branch scans, ticket checks, failed deliveries, refunds and operator statements. Require sign-off from passenger operations, logistics, finance, support and the technical owner. Release only when staff can resolve failures through documented workflows.

24 Implementation roadmap and staffing

The complete scope is retained throughout the roadmap. Staging reduces the number of simultaneous operational risks and lets later modules use foundations already tested in service.

Phase	Deliverables	Exit condition
1 Discovery and design	Operating model, service catalog, journeys, data rules, UX prototypes, provider onboarding	Approved workflows, prioritized backlog and integration access plan
2 Shared foundation	Identity, organizations, fleet, pricing, orders, permissions, audit, payment adapter	Core transactions and isolation tests pass
3 Passenger services	Scheduled booking, segment seats, manifests, rides, corporate approvals and charter	Each passenger model completes its main and exception journeys
4 Goods and hubs	Courier, packages, freight, line-haul, custody, warehouse and returns	Packages reconcile from collection to final release
5 Operational and financial controls	Reconciliation, refunds, settlement, maintenance, incidents and reporting	Operations and finance approve their daily closing process
6 Hardening and rollout	Load tests, security tests, restore drills, field pilot and gradual expansion	Launch gates pass and support ownership is active

Planning envelope

For the broad scope described here, use an initial planning envelope of roughly 8 to 14 months with an experienced multidisciplinary team, then replace it with a bottom-up estimate after discovery. A limited production pilot can arrive earlier. This is an estimate, not a supplier quote or delivery promise; integrations, hardware, regulatory approvals and custom warehouse needs can extend it.

Team responsibilities

A product owner sets business priorities. An operations specialist validates passenger and logistics workflows. A designer handles customer and staff interfaces. Backend engineers own transaction rules, data and integrations. Mobile engineers own field behavior and offline synchronization. Web engineers build portals. QA owns failure scenarios and acceptance evidence. Infrastructure/security ownership and a finance lead are required throughout, even if initially part-time.

Maintain a requirement-to-test traceability list, architecture decisions, API specifications, data migrations, operating runbooks and release notes. Train dispatch, branch and support teams before each service becomes available to customers.

25 Costs operating decisions and launch checklist

Cost model

Build a monthly cost model from actual usage instead of assuming one fixed hosting bill. Separate development cost, operational staffing and third-party usage. Current supplier quotes are needed before approving a budget.

Cost category	Main drivers
Engineering and design	Team size, delivery duration, quality work and ongoing maintenance
Cloud services	API/worker capacity, database, backups, storage, logging and data transfer
Maps	Map displays, route estimates, geocoding and optimization requests
Messaging	OTP, booking messages, delivery alerts and support traffic
Payments	Approved provider pricing, collection, payout and refund operations
Devices and connectivity	Driver phones, scanners, printers, trackers, SIM data and branch internet
Operations	Dispatch, branch handling, customer support, training and incident coverage
Compliance and assurance	Applicable permissions, insurance, privacy work, accounting and security review

Decisions to settle before contracts and implementation

Confirm whether the business is the carrier, a booking agent, a marketplace, or a combination by service. Identify who receives customer money, who issues invoices, who owes refunds and who bears claims. Choose initial operating locations, supported vehicle classes, cancellation rules, cargo acceptance policy, operator commission agreements and support hours.

Launch checklist

• Verify operator, vehicle and driver approvals; confirm applicable operating permissions and insurance arrangements.

• Complete production payment onboarding, reconciled trial transactions and refund/payout rehearsals.

• Publish accurate customer terms, privacy information, service prices and cancellation rules.

• Train staff on manifests, package scans, incidents, failed deliveries and finance closing.

• Pass authorization, concurrency, offline, security and backup-restoration tests.

• Assign alert owners, escalation contacts and a daily review of unresolved operational and payment exceptions.

The first launch can cover a controlled area or route set while the complete architecture remains available for expansion. Each added service should have its own operational owner and acceptance evidence.

26 References and glossary

Official product and technical sources used in this blueprint. Research date 18 September 2026. Product features and access terms may change; recheck integration documentation during implementation.

1 Little Mobility delivery and corporate services

2 BuuPass Bus routes operators and booking

3 Onfleet Delivery evidence options

3 Onfleet overview Dispatch route planning and delivery workflows

4 Samsara Vehicle tracking and fleet operations

5 Uber Freight Transportation management capabilities

6 Oracle Shipment and transport planning

7 Fleetio Maintenance and inspection workflows

8 Odoo Barcode receipts and deliveries

9 Safaricom M-PESA integration and API categories

10 KRA eTIMS system integration

11 Google Maps Route optimization objectives and constraints

12 Android Offline architecture

12 Android background location Location and battery constraints

13 AWS Transactional outbox and duplicate events

14 PostgreSQL Database locking

15 Flutter Mobile development documentation

16 NestJS Backend framework documentation

17 ODPC Data-controller and processor registration

18 Kenya Law 2022 transport-network-company regulations

Plain language glossary

Term	Meaning
API	A controlled way for applications to ask a system to perform work.
Backend	The server software that enforces rules and stores business records.
Idempotency	Repeating the same request safely produces one business effect.
Webhook	A provider sends your server an event such as a payment result.
Ledger	An auditable accounting record of money movements and obligations.
Custody	Who is responsible for a physical package at a particular time.
RPO and RTO	Recovery point measures potential data loss; recovery time measures restoration time.